

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			The {heat / energy} required to {raise / change} [the temperature of] 1 kg (1) by 1 °C or K (1)	2			2		
	(b)	(i)		Mass of the water = density $\times$ volume = $10^3 \times (1.2 \times 10^{-3}) = 1.2$ kg (1) Heat required to increase temperature = $shc \times \text{mass} \times \text{increase in temperature} = 4200 \times 1.2 \times (100 - 18) = [413\,280 \text{ J}]$ (1) Time required = $\frac{\text{heat required}}{\text{power}} = \frac{413\,280}{3 \times 10^3} = 137.8$ [s] (1) Or 2 min 18 s		3		3	3	
		(ii)		Use of density equation on milk [expect 0.037 kg] (1) Attempt at conservation of energy (1) e.g. Heat lost by the tea = Heat gained by the milk [expect 11 424 and 11 550 [J]] Correct equation set up (1) e.g. $(95 - 84)(4200)(0.25) = (84 - 5)(3900)m_{\text{milk}}$ or $(95 - \theta)(4200)(0.25) = (\theta - 5)(3900)m_{\text{milk}}$ or equivalent equation set up to calculate one of the variables or $0.25 \times 4200 \times 95 + 0.03708 \times 3900 \times 5 = (0.25 \times 4200 + 0.03708 \times 3900) \times T$ (note that this line also works if temperature is in K) Final answer and comment (1) e.g. $V = 3.64 \times 10^{-5} \text{ m}^3$ and so about right or final temperature = 84.1 °C and so about right						
				Question 4 total	2	3	4	9	6	0